

Prakhar Shekhar Parthasarathi

Full-Stack Software Engineer / Java & Spring Boot Backend / React Front-End / AWS

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🐙 github.com/praxstack · 🌐 linkedin.com/in/prakharshekhar ·

🌐 Portfolio · prax-portfolio-one.vercel.app

47%

P90 LATENCY CUT ·
BOOKING SURFACES

+33%

CONVERSION LIFT

195k rps

GET · JAVA REDIS FROM
SCRATCH

20→5

ON-CALL TICKETS /
SHIFT

≡ SUMMARY

Full-stack software engineer with **3+ years at Amazon** across Payments and Travel — **owned 4 microservices end-to-end** through design, build, deploy, and on-call. Java-heavy backend (**Java 17 / Spring Boot**, JVM concurrency with `CompletableFuture`, bounded `ExecutorService`, `ConcurrentHashMap`), idempotent event-driven architecture over **SQS + Step Functions**, and typed infrastructure in **AWS CDK (Java)** — paired with **React (SSR & CSR) + Redux** for customer-facing flows. Customer-facing impact: **47% p90** reduction on booking surfaces, **33% conversion** lift, **43 bps CPT** reduction via idempotent schedule-change automation, on-call tickets cut **20+ to ~5** per shift.

⌘ CORE TECHNICAL STACK

 JAVA & JVM Java 17 , Spring Boot, Spring MVC, Spring DI, Google Guice, JUnit 5, Mockito, SLF4J, Lombok, Maven, Gradle · <code>ExecutorService</code> , <code>CompletableFuture</code> , <code>ConcurrentHashMap</code> , thread-pool tuning, JVM profiling, GC analysis	 DISTRIBUTED SYSTEMS Microservices, REST, GraphQL, event-driven architecture, idempotency & dedup keys, state machines, retry with exponential backoff, DLQ handling, at-least-once semantics, TDD
 AWS Lambda (Java), DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, CDK (Java) — typed IaC for all service infrastructure	 FRONT-END React.js (SSR & CSR) , Redux (1.5 yrs production) — customer-facing Amazon flows; HTML / CSS, JavaScript / TypeScript
 DATA & STORAGE DynamoDB, Redis, NoSQL modelling, SQL, caching strategies	 OTHER LANGUAGES Python, C++, VTL
 TOOLING & PROCESS Git, GitHub, CI/CD, Gradle JaCoCo coverage gates, Agile / Scrum, on-call, code review, incident response	

Software Development Engineer

Jul 2022 – Sep 2025

📍 Amazon · Travel (Hotels & Flights) · Bengaluru

- **Service ownership — Spring Boot, Spring MVC.** Owned 4 **customer-facing microservices** powering the hotel booking platform — search, listing, room selection, order review — across design, code review, deploy, and on-call. Built REST controllers, service-layer logic, and `@Repository` DynamoDB DAOs wired through Spring DI. Shipped Prime-exclusive discount flow and partner-discount integration with tuned cache logic — **~30 bps** drop in funnel churn.
- **JVM concurrency — CompletableFuture, bounded executors.** Refactored sequential downstream calls into `CompletableFuture.allOf()` on a bounded `ExecutorService`; parallelised 6 partner-API lookups — hotel landing p90 cut **47%** (**1.1s → 0.59s**), conversion up **+33%**. GZIP on 150–200 MB catalog payloads cut wire size **95%**, eliminating network-timeout failures.
- **Event-driven — SQS + Step Functions.** Designed idempotent Java webhook for real-time flight schedule changes (cancellations, date-changes, non-operational flights) — SQS ingestion, DynamoDB-backed dedup keys, audit logging, automated offline modifications — addressing the highest-impact customer-pain category (**43 bps CPT**). Orchestrated end-to-end booking state machine (confirm / cancel / retry) on Step Functions with Java Lambda handlers.
- **Infrastructure as code — AWS CDK (Java).** Authored all service IaC in **Java CDK** — Lambdas, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch alarms. Migrated 2 services from hand-rolled CloudFormation to typed CDK stacks — compile-time infrastructure checks, **halved** rollback incidents.
- **Testing — JUnit 5, Mockito, Spring Boot.** Held **80%+** line coverage; Mockito argument captors for downstream contract verification; `@SpringBootTest` integration tests against in-memory DynamoDB. Introduced Gradle JaCoCo coverage gates in CI.
- **Operational excellence & high-bar automation.** Dove deep into multi-year on-call corpus — tickets cut from **20+ to ~5 per shift** (**~2 SDE-weeks / week** reclaimed). Java email-automation service deprecated stale WebLab experiments — config-hygiene incidents from 5–6 to **0** per on-call. Resolved **150+** production incidents; authored COEs and drove action items to closure. Built auto-insurance purchase flow (Java + React) with milestone persistence and auto-fill — **60%+** lift in renewals.

Software Development Engineer, Intern

May 2021 – Jul 2021 · Feb 2022 – Jul 2022

📍 Amazon · Bengaluru

- Migrated legacy GraphQL resolvers to **Java** — rebuilt resolver classes, wired them to DynamoDB via Spring-injected DAOs, exposed through API Gateway + Lambda.
- Shipped **One-Click Renewal** for auto-insurance using past-purchase data — async Java pre-fill workflow that materially simplified the funnel.
- Applied OO design and **Google Guice DI** throughout; landed a clean, testable codebase with **90%+** unit-test coverage (JUnit 5 + Mockito).

📁 SELECTED PROJECTS

AI Visual Code Review

github.com/praxstack/ai-visual-code-review · npm

Node.js Express TypeScript Jest VS Code API

Security-hardened visual code-review tool with a GitHub-style diff interface, real-time Git status tracking, AI-optimized markdown exports for ChatGPT / Claude, command-injection protection, and rate limiting. Published to npm and shipped as a VS Code extension.

Published: **npm v2.4.1** · MIT · CLI + VS Code extension.

Redis Server in Java

github.com/praxstack/redis-server-java

Java 17 Maven JUnit 5 Mockito

ConcurrentHashMap ScheduledExecutorService

RESP2-compatible Redis server built from scratch. Thread-per-client on a bounded **ExecutorService**; lock-free reads via **ConcurrentHashMap**; hybrid TTL eviction (lazy check on **GET** plus active **ScheduledExecutor** sweep); graceful shutdown via **Runtime.addShutdownHook**; integration tests over real TCP sockets.

Benchmarks (redis-benchmark, 100 concurrent clients):
SET **154k rps** · GET **195k rps** · INCR **128k rps**
· p50 < **0.3 ms** · **42 tests, 100% pass.**

Markdown Viewer App

github.com/praxstack/markdown-viewer-app

Vite Vanilla JS Marked.js PrismJS Mermaid

Vitest

Production-ready markdown viewer with real-time preview, syntax highlighting, Mermaid diagram rendering, 10 professional themes, and PDF / HTML export. Architected with modular vanilla JavaScript — zero framework overhead, sub-100 KB bundle.

Metrics: **155+** Vitest tests · real-time preview · Mermaid diagrams · PDF / HTML export.

Coach Atlas

github.com/praxstack/coach-atlas

React TypeScript Vite Bedrock Mermaid

KaTeX

An AI technical-interview mentor and content studio — Socratic interview prep, production-grade tutorial generation, and Mermaid / KaTeX / code rendering. Multi-provider (OpenAI, Anthropic, Gemini, AWS Bedrock) with a BYOK design: keys live in the browser, never on a server.

Stack: React + TypeScript · multi-provider BYOK · deployed on Vercel.

Open-Source Contributions

Merged into **danielmiessler/Fabric** (Go, 42k★ AI-tooling framework) — three production PRs:

#2044 — added AWS Bedrock bearer-token (ABSK) authentication + guided **--setup** (the same auth Claude Code uses).
+882 / -83

#2052 — dynamic Bedrock region fetching from botocore (40+ regions) + AWS_PROFILE conflict fix. **+252 / -15**

#2061 — fixed a streaming-goroutine deadlock in **Chatter.Send** + unified CLI/REST strategy handling. **+494 / -42**

Open PRs under review: **bytedance/deer-flow** (75k★ · model-config normalization), **thedotmack/claude-mem** (85k★ · plugin packaging), **refactoringhq/tolaria** (17k★ · editor callouts).

🎓 EDUCATION

M.C.A., Computer Science · Motilal Nehru National Institute of Technology (MNNIT), Prayagraj **GPA 9.25** **2019 – 2022**

B.Sc. (Hons), Computer Science · Bhaskaracharya College of Applied Sciences, University of Delhi **2015 – 2018**

🏆 AWARDS & RECOGNITION

Amazon Pay Merchants Categories Champion — 100+ hotel trust-buster bugs resolved; on-call tickets cut **20+ to ~5** per shift. **2023**